2025-2026 NC Math 3 Scope and Sequence (Standard & Honors)

High School Semester Block

Refer to Unit Planning Organizers for Instructional Guidance

Unit	Unit Title	S1: ≈ # of Weeks	S1: Date Range	S2: ≈ # of Weeks	S2: Date Range
1	***Incorporate WIM - Week of Inspirational Math*** Functions & Their Inverses	2 weeks	8/13 - 8/26	2 weeks	1/5 - 1/16
2	Exponential & Logarithmic Functions	≈2 weeks	8/27 - 9/9	≈2 weeks	1/20 - 1/30
3	Polynomial Functions	2 weeks	9/10 - 9/23	2 weeks	2/2 - 2/13
4	Rational Functions	2 weeks	9/24 - 10/7	2 weeks	2/16 - 2/27
	DPI NC Check-In 1	***Suggested after Unit 4***			
5	Modeling w/Geometry	≈2 weeks	10/8 - 10/21	2 weeks	3/2 - 3/20
6	Reasoning w/Geometry with Circles	≈3 weeks	10/22 - 11/7	2 weeks	3/23 - 4/10
	DPI NC Check-In 2	***Suggested after Unit 6***			
7	Trigonometric Functions	≈2 weeks	11/10 - 11/21	≈2 weeks	4/13 - 4/24
8	Statistics	≈1 ½ weeks	11/24 - 12/5	≈1 week	4/27 - 5/1
	Review **EOC**		**12/8 - 12/12** **12/15 – 12/19**		**5/4 – 5/8** **5/11 – 5/22**

The number of suggested weeks for each unit includes assessment and review days.
Each unit will need flexibility based on schedules, etc.

Standards for Mathematical Practice	
1. Make sense of problems and persevere in solving them.	5. Use appropriate tools strategically.
2. Reason abstractly and quantitatively.	6. Attend to precision.
3. Construct viable arguments and critique reasoning of others.	7. Look for and make use of structure.
4. Model with mathematics	8. Look for and express regularity in repeated reasoning.

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# Unit 1: Functions and their Inverses	≈ # of Weeks: S1: 2 Week / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Interpret functions that arise in applications in terms of the context.
- Analyze functions using different representations.
- Build a function that models a relationship between two quantities.
- Build new functions from existing functions.
- Create equations that describe numbers or relationships.
- Interpret the structure of expressions.
- Represent and solve equations and inequalities graphically.

SMPs: *See Unit Planning Organizer*

Interpreting Functions	
NC.M3.F-IF.2	Use function notation to evaluate piecewise defined functions for inputs in their domains, and interpret statements that use function notation in terms of a context.
NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.
NC.M3.F-IF.7	Analyze **piecewise, absolute value**, **polynomials (quadratics)**, exponential, rational, and trigonometric functions (sine and cosine) using different representations to show key features of the graph, by hand in simple cases and using technology for more complicated cases, including: **domain and range**; **intercepts**; **intervals where the function is increasing**, **decreasing**, **positive**, or **negative**; **rate of change**; **relative maximums and minimums**; symmetries; end behavior; period; and discontinuities.
NC.M3.F-IF.9	Compare key features of two functions using different representations by comparing properties of two different functions,

	each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
Building Functions	
NC.M3.F-BF.1b	Write a function that describes a relationship between quantities. b. Build a new function, in terms of a context, by combining standard function types using arithmetic operations.
NC.M3.F-BF.3	Extend an understanding of the effects on the graphical and tabular representations of a function when replacing $f(x)$ with $k{\cdot}f(x)$, $f(x)+k$, $f(x+k)$ to include $f(k{\cdot}x)$ for specific values of k (both positive and negative).
NC.M3.F-BF.4 NC.M3.F-BF.4a NC.M3.F-BF.4b NC.M3.F-BF.4c	Find an inverse function. a. Understand the inverse relationship between exponential and logarithmic, **quadratic and square root**, and **linear to linear functions** and use this relationship to solve problems using tables, graphs, and equations. b. Determine if an inverse function exists by analyzing tables, graphs, and equations. *c.* If an inverse function exists for a **linear**, **quadratic** and/or exponential function, f, represent the inverse function, f^{-1}, with a table, graph, or equation and use it to solve problems in terms of a context.
Creating Equations	
NC.M3.A-CED.1	Create equations and inequalities in one variable that represent **absolute value**, **polynomial (quadratics)**, exponential, and rational relationships and use them to solve problems algebraically and graphically.
NC.M3.A-CED.2	Create and graph equations in two variables to represent **absolute value**, **polynomial (quadratics)**, exponential and rational relationships between quantities.
NC.M3.A-CED.3	Create systems of equations and/or inequalities to model situations in context.
Seeing Structure in Expressions	
NC.M3.A-SSE.1 NC.M3.ASSE.1a	Interpret expressions that represent a quantity in terms of its context. a. Identify and interpret parts of a piecewise, absolute value, polynomial (quadratics), exponential and rational expressions

NC.M3.ASSE.1b	including terms, factors, coefficients, and exponents. b. Interpret expressions composed of multiple parts by viewing one or more of their parts as a single entity to give meaning in terms of a context.
Reasoning with Equations and Inequalities	
NC.M3.A-REI.1	Justify a solution method for equations and explain each step of the solving process using mathematical reasoning.
NC.M3.A-REI.11	Extend an understanding that the x-coordinates of the points where the graphs of two equations y = f(x) and y = g(x) intersect are the solutions of the equation $f(x) = g(x)$ and approximate solutions using a graphing technology or successive approximations with a table of values.

# Unit 2: Exponential and Logarithmic Functions	**≈ # of Weeks:** **S1: 2 Weeks / S2: 2 Weeks**

Concepts to Integrate: Apply all to real world applications.

- Create exponential or logarithmic equations that describe numbers or relationships.
- Interpret the structure of exponential or logarithmic expressions.
- Write exponential or logarithmic expressions in equivalent forms to solve problems.
- Build new functions from existing exponential or logarithmic functions.
- Interpret exponential or logarithmic functions that arise in applications in terms of the context.
- Analyze exponential or logarithmic functions using different representations.
- Build an exponential or logarithmic function that models a relationship between two quantities.
- Build new exponential or logarithmic functions from existing functions.
- Construct and compare linear and exponential models and solve problems.

SMPs: *See Unit Planning Organizer*

Creating Equations	
NC.M3.A-CED.1	Create and graph equations in one variable to represent absolute value, polynomial, **exponential** and rational relationships between quantities.
NC.M3.A-CED.2	Create and graph equations in two variables to represent absolute value, polynomial, **exponential** and rational relationships between quantities.

Reasoning with Equations and Inequalities

Understand solving equations as a process of reasoning and explain the reasoning

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NC.M3.A-REI.1	Justify a solution method for equations and explain each step of the solving process using mathematical reasoning.
Algebra	
NC.M3.A-SSE.1 NC.M3.A-SSE.1a NC.M3.A-SSE.1b	Interpret expressions that represent a quantity in terms of its context. a. Identify and interpret parts of a piecewise, absolute value, polynomial, **exponential** and rational expressions including terms, factors, coefficients, and exponents. b. Interpret expressions composed of multiple parts by viewing one or more of their parts as a single entity to give meaning in terms of a context.
NC.M3.A-SSE.2	Use the structure of an expression to identify ways to write equivalent expressions.
Seeing Structure in Expressions	
NC.M3.A-SSE.3c	Write an equivalent form of an exponential expression by using the properties of exponents to transform expressions to reveal rates based on different intervals of the domain.
Building Functions	
NC.M3.F-BF.1 NC.M3.F-BF.1a NC.M3.F-BF.1b	Write a function that describes a relationship between quantities. a. Build polynomial and **exponential** functions with real solution(s) given a graph, a description of a relationship, or ordered pairs (include reading these from a table). b. Build a new function, in terms of a context, by combining standard function types using arithmetic operations.
NC.M3.F-BF.3	Extend an understanding of the effects on the graphical and tabular representations of a function when replacing $f(x)$ with $k{\cdot}f(x)$, $f(x)+k$, $f(x+k)$ to include $f(k{\cdot}x)$ for specific values of k (both positive and negative).

NC.M3.F-BF.4 NC.M3.F-BF.4a NC.M3.F-BF.4b NC.M3.F-BF.4c	Find an inverse function. a. Understand the inverse relationship between **exponential and logarithmic**, quadratic and square root, and linear to linear functions and use this relationship to solve problems using tables, graphs, and equations. b. Determine if an inverse function exists by analyzing tables, graphs, and equations. *c.* If an inverse function exists for a linear, quadratic and/or **exponential function**, f, represent the inverse function, f^{-1}, with a table, graph, or equation and use it to solve problems in terms of a context.
Interpreting Functions	
NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.
NC.M3.F-IF.7	Analyze piecewise, absolute value, polynomials, **exponential**, rational, and trigonometric functions (sine and cosine) using different representations to show key features of the graph, by hand in simple cases and using technology for more complicated cases, including: **domain and range**; **intercepts**; **intervals where the function is increasing, decreasing, positive, or negative**; rate of change; relative maximums and minimums; symmetries; end behavior; period; and discontinuities.
NC.M3.F-IF.9	Compare key features of two functions using different representations by comparing properties of two different functions, each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
Linear, Quadratic, and Exponential Models	
NC.M3.F-LE.4	Use logarithms to express the solution to $ab^{ct} = d$ where a, c, and d are numbers and evaluate the logarithm using technology

Unit 3: Polynomial Functions

≈ # of Weeks:
S1: 2 Weeks / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Explain volume formulas and use them to solve problems.
- Apply geometric concepts in modeling situations.
- Use complex numbers in polynomial identities and equations.
- Interpret the structure of polynomial expressions.
- Understand the relationship between zeros and factors of polynomials.
- Create polynomial equations that describe numbers or relationships.
- Interpret polynomial functions that arise in applications in terms of the context.
- Build a polynomial function that models a relationship between two quantities.
- Build new functions from existing functions.
- Construct and compare linear and exponential models and solve problems.

SMPs: *See Unit Planning Organizer*

Geometric Measure and Dimension	
NC.M3.G-GMD.3	Use the volume formulas for prisms, cylinders, pyramids, cones, and spheres to solve problems.
Modeling with Geometry	
NC.M3.G-MG.1	Apply geometric concepts in modeling situations • Use geometric and algebraic concepts to solve problems in modeling situations: • Use geometric shapes, their measures, and their properties, to model real-life objects. • Use geometric formulas and algebraic functions to model relationships. • Apply concepts of density based on area and volume.

	• Apply geometric concepts to solve design and optimization problems.
The Complex Number System	
NC.M3.N-CN.9	Use the Fundamental Theorem of Algebra to determine the number and potential types of solutions for polynomial functions.
Seeing Structure in Expressions	
NC.M3.A-SSE.1 NC.M3.A-SSE.1a NC.M3.A-SSE.1b	Interpret expressions that represent a quantity in terms of its context. a. Identify and interpret parts of a piecewise, absolute value, **polynomial**, exponential and rational expressions including terms, factors, coefficients, and exponents. b. Interpret expressions composed of multiple parts by viewing one or more of their parts as a single entity to give meaning in terms of a context.
NC.M3.A-SSE.2	Use the structure of an expression to identify ways to write equivalent expressions.
Arithmetic with Polynomial and Rational Expressions	
NC.M3.A-APR.2	Understand and apply the Remainder Theorem.
NC.M3.A-APR.3	Understand the relationship among factors of a polynomial expression, the solutions of a polynomial equation and the zeros of a polynomial function.
Creating Equations	
NC.M3.A-CED.1	Create equations and inequalities in one variable that represent absolute value, **polynomial**, exponential, and rational relationships and use them to solve problems algebraically and graphically.
NC.M3.A-CED.2	Create and graph equations in two variables to represent absolute value, **polynomial**, exponential and rational relationships between quantities.

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Reasoning with Equations and Inequalities	
NC.M3.A-REI.1	Justify a solution method for equations and explain each step of the solving process using mathematical reasoning.
Interpreting Functions	
NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.
NC.M3.F-IF.7	Analyze piecewise, absolute value, **polynomials**, exponential, rational, and trigonometric functions (sine and cosine) using different representations to show key features of the graph, by hand in simple cases and using technology for more complicated cases, including: **domain and range**; **intercepts; intervals where the function is increasing, decreasing, positive, or negative**; rate of change; **relative maximums and minimums**; **symmetries; end behavior**; period; and discontinuities.
NC.M3.F-IF.9	Compare key features of two functions using different representations by comparing properties of two different functions, Each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
Building Functions	
NC.M3.F-BF.1 NC.M3.F-BF.1a NC.M3.F-BF.1b	Write a function that describes a relationship between quantities. a. Build **polynomial** and exponential functions with real solution(s) given a graph, a description of a relationship, or ordered pairs (include reading these from a table). b. Build a new function, in terms of a context, by combining standard function types using arithmetic operations.
NC.M3.F-BF.3	Extend an understanding of the effects on the graphical and tabular representations of a function when replacing $f(x)$ with $k{\cdot}f(x)$, $f(x)+k$, $f(x+k)$ to include $f(k{\cdot}x)$ for specific values of k (both positive and negative).
Linear, Quadratic, and Exponential Models	
NC.M3.F-LE.3	Compare the end behavior of functions using their rates of change over intervals of the same length to show that a quantity increasing exponentially eventually exceeds a quantity increasing as a polynomial function.

Unit 4: Rational Functions

≈ # of Weeks:
S1: 2 Weeks / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Interpret the structure of rational expressions.
- Rewrite rational expressions.
- Create rational equations that describe numbers relationships.
- Understand solving rational equations as a process of reasoning and explain the reasoning.
- Interpret rational functions that arise in applications in terms of the context.

SMPs: *See Unit Planning Organizer*

Seeing Structure in Expressions	
NC.M3.A-SSE.1 NC.M3.A-SSE.1a NC.M3.A-SSE.1b	Interpret expressions that represent a quantity in terms of its context. a. Identify and interpret parts of a piecewise, absolute value, polynomial, exponential and **rational** expressions including terms, factors, coefficients, and exponents. b. Interpret expressions composed of multiple parts by viewing one or more of their parts as a single entity to give meaning in terms of a context.
Arithmetic with Polynomial and Rational Expressions	
NC.M3.A-APR.6	Rewrite simple rational expressions in different forms; write a(x)/b(x) in the form q(x) + r(x)/b(x), where a(x), b(x), q(x), and r(x) are polynomials with the degree of r(x) less than the degree of b(x).
NC.M3.A-APR.7	Understand the similarities between arithmetic with rational expressions and arithmetic with rational numbers. a. Add and subtract two rational expressions, $a(x)$ and $b(x)$, where the denominators of both $a(x)$ and $b(x)$ are linear expressions. b. Multiply and divide two rational expressions.

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Creating Equations	
NC.M3.A-CED.1	Create and graph equations in one variable to represent absolute value, polynomial, exponential and **rational** relationships between quantities.
NC.M3.A-CED.2	Create and graph equations in two variables to represent absolute value, polynomial, exponential and **rational** relationships between quantities.
Reasoning with Equations and Inequalities	
NC.M3.A-REI.1	Justify a solution method for equations and explain each step of the solving process using mathematical reasoning.
NC.M3.A-REI.2	Solve and interpret one variable rational equations arising from a context, and explain how extraneous solutions may be produced.
Interpreting Functions	
NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.
NC.M3.F-IF.7	Analyze piecewise, absolute value, polynomials, exponential, **rational**, and trigonometric functions (sine and cosine) using different representations to show key features of the graph, by hand in simple cases and using technology for more complicated cases, including: **domain and range**; **intercepts; intervals where the function is increasing, decreasing, positive, or negative**; rate of change; relative maximums and minimums; **symmetries**; end behavior; period; and **discontinuities**.
NC.M3.F-IF.9	Compare key features of two functions using different representations by comparing properties of two different functions, each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).

Unit 5: Modeling with Geometry

≈ # of Weeks:
S1: 2 Weeks / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Translate between the geometric description and the equation for a conic section.
- Explain volume formulas and use them to solve problems.
- Visualize relationships between two-dimensional and three-dimensional objects.
- Apply geometric concepts in modeling situations.
- Prove geometric theorems.

SMPs: ALL

Interpret the Structure of Expressions	
NC.M3.A-SSE.2	Use the structure of an expression to identify ways to write equivalent expressions.
Geometric Measurement and Dimension	
NC.M3.G-GMD.3	Use the volume formulas for prisms, cylinders, pyramids, cones, and spheres to solve problems.
NC.M3.G-GMD.4	Identify the shapes of two-dimensional cross-sections of three-dimensional objects, and identify three-dimensional objects generated by rotations of two-dimensional objects.
Modeling with Geometry	
NC.M3.G-MG.1	Apply geometric concepts in modeling situations ● Use geometric and algebraic concepts to solve problems in modeling situations: ● Use geometric shapes, their measures, and their properties, to model real-life objects. ● Use geometric formulas and algebraic functions to model relationships.

	• Apply concepts of density based on area and volume. • Apply geometric concepts to solve design and optimization problems.
Congruence	
NC.M3.G-CO.10	Verify experimentally properties of the centers of triangles (centroid, incenter, and circumcenter).
NC.M3.G-CO.11	Prove theorems about parallelograms. • Opposite sides of a parallelogram are congruent. • Opposite angles of a parallelogram are congruent. • Diagonals of a parallelogram bisect each other. • If the diagonals of a parallelogram are congruent, then the parallelogram is a rectangle.
NC.M3.G-CO.14	Apply properties, definitions, and theorems of two-dimensional figures to prove geometric theorems and solve problems.

Check in #1 Window	**# Days: 1**

Unit 6: Reasoning with Geometry (Circles)

≈ # of Weeks:
S1: 3 Weeks / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Understand and apply theorems about circles.
- Prove geometric theorems.
- Translate between the geometric description and the equation for a conic section.
- Apply geometric concepts in modeling situations.

SMPs: *See Unit Planning Organizer*

Interpret the structure of expressions.	
NC.M3.A-SSE.2	Use the structure of an expression to identify ways to write equivalent expressions.
Circles	
NC.M3.G-C.2	Understand and apply theorems about circles. • Understand and apply theorems about relationships with angles and circles, including central, inscribed and circumscribed angles. • Understand and apply theorems about relationships with line segments and circles including, radii, diameter, secants, tangents and chords.
NC.M3.G-C.5	Using similarity, demonstrate that the length of an arc, s, for a given central angle is proportional to the radius, r, of the circle. Define radian measure of the central angle as the ratio of the length of the arc to the radius of the circle, s/r. Find arc lengths and areas of sectors of circles.
Expressing Geometric Properties with Equations	
NC.M3.G-GPE.1	Derive the equation of a circle of given center and radius using the Pythagorean Theorem; complete the square to find the

	center and radius of a circle given by an equation.
Modeling with Geometry	
NC.M3.G-MG.1	Apply geometric concepts in modeling situations ● Use geometric and algebraic concepts to solve problems in modeling situations: ● Use geometric shapes, their measures, and their properties, to model real-life objects. ● Use geometric formulas and algebraic functions to model relationships. ● Apply concepts of density based on area and volume. ● Apply geometric concepts to solve design and optimization problems.

Unit 7: Trigonometric Functions

≈ # of Weeks:
S1: 2 Weeks / S2: 2 Weeks

Concepts to Integrate: Apply all to real world applications.

- Build new functions from existing trigonometric functions.
- Understand the concept of a trigonometric function and use function notation.
- Interpret trigonometric functions that arise in applications in terms of the context.
- Analyze trigonometric functions using different representations.
- Extend the domain of trigonometric functions using the unit circle.
- Model periodic phenomena with trigonometric functions.

SMPs: *See Unit Planning Organizer*

Algebra ***Overarching Standard***	
NC.M3.A-SSE.lb	Use the structure of an expression to identify ways to write equivalent expressions.
Building Functions	
NC.M3.F-BF.3	Extend an understanding of the effects on the graphical and tabular representations of a function when replacing $f(x)$ with $k{\cdot}f(x)$, $f(x)+k$, $f(x+k)$ to include $f(k{\cdot}x)$ for specific values of k (both positive and negative).
Interpreting Functions	
NC.M3.F-IF.1	Extend the concept of a function by recognizing that trigonometric ratios are functions of angle measure.

NC.M3.F-IF.4	Interpret key features of graphs, tables, and verbal descriptions in context to describe functions that arise in applications relating two quantities to include periodicity and discontinuities.
NC.M3.F-IF.7	Analyze piecewise, absolute value, polynomials, exponential, rational, and **trigonometric functions (sine and cosine)** using different representations to show key features of the graph, by hand in simple cases and using technology for more complicated cases, including: **domain and range; intercepts; intervals where the function is increasing, decreasing, positive, or negative**; rate of change; **relative maximums and minimums; symmetries**; end behavior; **period**; and discontinuities.
NC.M3.F-IF.9	Compare key features of two functions using different representations by comparing properties of two different functions, each with a different representation (symbolically, graphically, numerically in tables, or by verbal descriptions).
Trigonometric Functions	
NC.M3.F-TF.1	Understand radian measure of an angle as: ● The ratio of the length of an arc on a circle subtended by the angle to its radius. ● A dimensionless measure of length defined by the quotient of arc length and radius that is a real number. ● The domain for trigonometric functions.
NC.M3.F-TF.2	Build an understanding of trigonometric functions by using tables, graphs and technology to represent the cosine and sine functions. a. Interpret the sine function as the relationship between the radian measure of an angle formed by the horizontal axis and a terminal ray on the unit circle and its y coordinate. b. Interpret the cosine function as the relationship between the radian measure of an angle formed by the horizontal axis and a terminal ray on the unit circle and its x coordinate.
NC.M3.F-TF.5	Use technology to investigate the parameters, a, b, and h of a sine function, $f(x) = a \cdot sin(b \cdot x) + h$, to represent periodic phenomena and interpret key features in terms of a context.

Check in #2 Window	**# Days: 1**

Unit 8: Statistics	**≈ # of Weeks:** **S1: 1 Week / S2: 1 Week**
Review/NC Math 3 EOC	
Concepts to Integrate: Apply all to real world applications. ● Understand and evaluate random processes underlying statistical experiments. ● Make inferences and justify conclusions from sample surveys, experiments, and observational studies.	
SMPs: ALL	
Algebra ***Overarching Standard***	
NC.M3.A-SSE.lb	Use the structure of an expression to identify ways to write equivalent expressions.
Making Inference and Justifying Conclusions	
NC.M3.S-IC.1	Understand the process of making inferences about a population based on a random sample from that population.
NC.M3.S-IC.3	Recognize the purposes of and differences among sample surveys, experiments, and observational studies and understand how randomization should be used in each.
NC.M3.S-IC.4	Use simulation to understand how samples can be used to estimate a population mean or proportion and how to determine a margin of error for the estimate.

NC.M3.S-IC.5	Use simulation to determine whether observed differences between samples from two distinct populations indicate that the two populations are actually different in terms of a parameter of interest.
NC.M3.S-IC.6	Evaluate articles and websites that report data by identifying the source of the data, the design of the study, and the way the data are graphically displayed.